```
mysql> SELECT 'DeptID is the primary key of Department.';
+-----------------------------------------------+
| DeptID is the primary key of Department.      |
+-----------------------------------------------+
| DeptID uniquely identifies each department.   |
+-----------------------------------------------+
1 row in set (0.00 sec)


mysql> CREATE DATABASE IF NOT EXISTS sql_lab;
Query OK, 1 row affected (0.01 sec)

mysql> USE sql_lab;
Database changed

mysql> CREATE TABLE Department (
    -> DeptID INT PRIMARY KEY,
    -> DeptName VARCHAR(50) NOT NULL
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Department VALUES
    -> (10,'Computer Science'),
    -> (20,'Information Technology'),
    -> (30,'Pharmacy'),
    -> (40,'Management');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Department;
+--------+----------------------+
| DeptID | DeptName             |
+--------+----------------------+
|     10 | Computer Science     |
|     20 | Information Technology|
|     30 | Pharmacy             |
|     40 | Management           |
+--------+----------------------+
4 rows in set (0.00 sec)


mysql> CREATE TABLE Student (
    -> StudentID INT PRIMARY KEY,
    -> Name VARCHAR(100) NOT NULL,
    -> Email VARCHAR(100),
    -> Age INT,
    -> DeptID INT,
    -> JoiningDate DATE
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Student VALUES
    -> (101,'Amit Sharma','amit@example.com',20,10,'2026-07-01'),
    -> (102,'Priya Singh','priya@example.com',21,20,'2026-07-03'),
    -> (103,'Rahul Verma','rahul@example.com',22,30,'2026-07-05'),
    -> (104,'Neha Gupta','neha@example.com',20,10,'2026-07-08'),
    -> (105,'Arjun Mehta','arjun@example.com',23,40,'2026-07-10');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Student;
+-----------+-------------+-------------------+------+--------+-------------+
| StudentID | Name        | Email             | Age  | DeptID | JoiningDate |
+-----------+-------------+-------------------+------+--------+-------------+
|       101 | Amit Sharma | amit@example.com  |   20 |     10 | 2026-07-01  |
|       102 | Priya Singh | priya@example.com |   21 |     20 | 2026-07-03  |
|       103 | Rahul Verma | rahul@example.com |   22 |     30 | 2026-07-05  |
|       104 | Neha Gupta  | neha@example.com  |   20 |     10 | 2026-07-08  |
|       105 | Arjun Mehta | arjun@example.com |   23 |     40 | 2026-07-10  |
+-----------+-------------+-------------------+------+--------+-------------+
5 rows in set (0.00 sec)


mysql> INSERT INTO Student
    -> VALUES (101,'Riya Sharma','riya@example.com',21,20,'2026-07-12');
ERROR 1062 (23000): Duplicate entry '101' for key 'student.PRIMARY'


mysql> CREATE TABLE Enrollment (
    -> EnrollmentID INT PRIMARY KEY,
```

```
    -> StudentID INT,
    -> CourseName VARCHAR(100)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Enrollment VALUES
    -> (1,101,'DBMS'),
    -> (2,102,'Operating Systems'),
    -> (3,103,'Pharmacology'),
    -> (4,104,'Python');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Enrollment;
+--------------+-----------+-------------------+
| EnrollmentID | StudentID | CourseName        |
+--------------+-----------+-------------------+
|            1 |       101 | DBMS              |
|            2 |       102 | Operating Systems |
|            3 |       103 | Pharmacology      |
|            4 |       104 | Python            |
+--------------+-----------+-------------------+
4 rows in set (0.00 sec)


mysql> INSERT INTO Enrollment VALUES (1,105,'AI');
ERROR 1062 (23000): Duplicate entry '1' for key 'enrollment.PRIMARY'


mysql> ALTER TABLE Student
    -> ADD CONSTRAINT fk_student_department
    -> FOREIGN KEY (DeptID) REFERENCES Department(DeptID);
Query OK, 5 rows affected (0.05 sec)
Records: 5  Duplicates: 0  Warnings: 0


mysql> ALTER TABLE Enrollment
    -> ADD CONSTRAINT fk_enrollment_student
    -> FOREIGN KEY (StudentID) REFERENCES Student(StudentID);
Query OK, 4 rows affected (0.04 sec)
Records: 4  Duplicates: 0  Warnings: 0


mysql> INSERT INTO Student
    -> VALUES (106,'Karan Joshi','karan@example.com',22,99,'2026-07-12');
ERROR 1452 (23000): Cannot add or update a child row:
a foreign key constraint fails (`sql_lab`.`student`,
CONSTRAINT `fk_student_department` FOREIGN KEY (`DeptID`)
REFERENCES `department` (`DeptID`))


mysql> INSERT INTO Enrollment VALUES (5,999,'Artificial Intelligence');
ERROR 1452 (23000): Cannot add or update a child row:
a foreign key constraint fails (`sql_lab`.`enrollment`,
CONSTRAINT `fk_enrollment_student` FOREIGN KEY (`StudentID`)
REFERENCES `student` (`StudentID`))


mysql> SELECT s.StudentID, s.Name, d.DeptName
    -> FROM Student s
    -> INNER JOIN Department d ON s.DeptID = d.DeptID;
+-----------+-------------+-----------------------+
| StudentID | Name        | DeptName              |
+-----------+-------------+-----------------------+
|       101 | Amit Sharma | Computer Science      |
|       102 | Priya Singh | Information Technology|
|       103 | Rahul Verma | Pharmacy              |
|       104 | Neha Gupta  | Computer Science      |
|       105 | Arjun Mehta | Management            |
+-----------+-------------+-----------------------+
5 rows in set (0.00 sec)


mysql> SELECT e.EnrollmentID, e.StudentID, s.Name, e.CourseName
    -> FROM Enrollment e
    -> INNER JOIN Student s ON e.StudentID = s.StudentID;
+--------------+-----------+-------------+-------------------+
| EnrollmentID | StudentID | Name        | CourseName        |
+--------------+-----------+-------------+-------------------+
|            1 |       101 | Amit Sharma | DBMS              |
```

```
|            2 |       102 | Priya Singh | Operating Systems |
|            3 |       103 | Rahul Verma | Pharmacology      |
|            4 |       104 | Neha Gupta  | Python            |
+--------------+-----------+-------------+-------------------+
4 rows in set (0.00 sec)
```